



Faculty of Engineering
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Menstrual Health & Wellness Journal: Designed for Cycle Complexity and Meaningful Engagement

Design Team

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Abstract

This paper presents the *Menstrual Health & Wellness Journal*, a user-centered tool that fills the gap left by impersonal apps and generic journals by supporting people who menstruate in tracking, understanding, and reflecting on their menstrual health with agency and flexibility. Building on prior work by Mouradyan, A., Pappu, N., Naufal, S., & Thompson, S [1], this prototype was developed through an iterative participatory process involving user interviews, focus groups, and competitive market analysis, incorporating feedback from individuals aged 18 to 30 who menstruate. The prototype evolved through three major iterations based on stakeholder feedback, each refining usability, clarity, and user alignment. Core features include customizable symptom tracking, phase-specific educational content, reflective prompts, and tactile interactive elements designed for daily engagement. Aesthetically, the journal adopts the minimalist structure and visual clarity of popular reflection journals, filling a critical market gap often left by existing menstrual health journals.

While the current prototype effectively meets the needs of users seeking a reflective, low-tech alternative to digital tools, limitations remain in terms of long-term validation, product sophistication, and inclusivity of diverse menstrual experiences. Future work will explore extended diary studies and potential distribution as a printable resource to promote equitable access. Ultimately, this project contributes a novel, physical design solution to an underrepresented area of wellness, offering users a more personalized and validating relationship with their menstrual cycle.

1. Design Overview

Menstrual health is a complex and often stigmatized area within healthcare and broader society. Leveraging past user-centered design requirements centered around physical, emotional, and social factors [1], this proposed solution centers on the development of a journal. The *Menstrual Health & Wellness* journal is an analog tool designed to support users, track, learn, and understand their own menstrual health and overall wellbeing.

Previous contextual inquiry and participatory design sessions were conducted to understand the experience of individuals who menstruate amounting to a set of core design requirements (Table 1), as described more thoroughly in Section 2.3. The two requirements most consistently ranked as important were:

1. One's own health is one's own responsibility (requirement #1)
2. Menstrual cycles are unique, and regularity is subjective (requirement #5)

The journal design presented in Figure 1 and Appendix (Section 11) addresses the basic need for users to take an active responsibility for their menstrual health while accounting for uniqueness and flexibility. Users were engaged in semi-structured interviews and focus groups throughout the prototype development process, as described in Section 3.2, to reliably incorporate their needs into the design. The manner in which the proposed prototype meets user needs is expanded on in Section 4 of this design report.

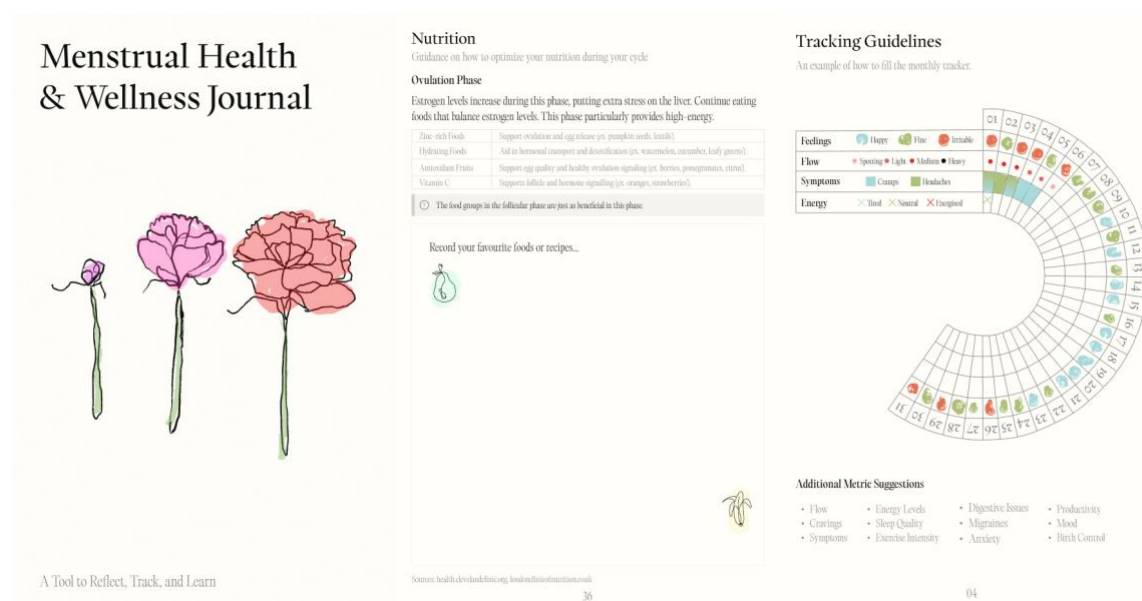


Figure 1. Sample Pages From Prototype

The journal content includes:

Pages 1 - 3: Opening Notes & Introduction

Illustrations are leveraged to draw user attention to the cover, and provide context on the purpose and intention behind this journal. Additionally, the opening notes emphasise the variability in cycles, and to consult professional medical support as needed.

Page 4: Goal Setting Template

This is a place for intentionally setting goals related to different facets of menstrual health and wellness, including nutrition, exercise, emotional health, rest, education, and lifestyle.

Pages 5 - 30: Monthly Tracking & Reflection Pages

Users are provided detailed instructions and tracking examples showcasing various symptoms and how to leverage multimedia (stickers, paint, pens, icons etc.). They're provided a tracking template for each month, alongside a reflection page per month.

Pages 31 - 34: Cycle Phases

Users are provided information on each phase of the menstrual cycle (luteal, menstruation, ovulation, and follicular) as well as an interactive wheel to understand how educational material is translated to their cycle.

Pages 35 - 40: Nutrition

Users are given recommendations on types of foods to eat in each menstrual phase with rationale and a detachable food pyramid to guide meal planning.

Pages 41 - 44: Exercise

Similarly, exercise needs vary throughout the menstrual health cycle. Users are provided guidance on which exercises align with the constraints and opportunities of each cycle phase.

Pages 45 - 48: Postcards

Detachable postcards are provided with instructions on how to utilize them to either connect with other people or oneself.

Pages 49 - 51: Connection

Twelve prompts (roughly intended to be completed once per month) are provided to help users engage deeply with others or reflect. Prompts are designed to help understand and preserve lived experiences that friends and family have when navigating menstrual health.

Page 52: Question

Throughout this process of tracking and learning, users can jot down questions for their doctors, trusted information sources, the internet, etc.

Page 53: Resources

This journal is designed for timelessness and lightweight learning, but additional resources are provided for users to seek out more information. Users can interact with the list by adding more sources tailored to their needs.

2. Background

2.1. Problem Definition

Menstrual health is a dynamic challenge made complicated through the integration of biological, psychological, and societal factors. In previous work, an investigation was conducted to understand menstrual health-related challenges and identify relevant opportunities for improvement [1].

The full design process included observation, data collection, establishing design requirements and participatory design. All the involved stakeholders can be found in Table #1 in the Appendix (Section 0) of this report. Following a problem definition structure, the basic need discovered was to independently and securely monitor one's menstrual health while gaining increased autonomy over one's health [2]. Though this proposed design is an asset to all people who menstruate, it especially benefits those who:

- Have no existing way of monitoring their health
- Prefer physical over digital data due to data security
- Have minimal motivation to monitor their menstrual health (ie, apps can become tiresome)
- Seek customizability in their menstrual health tracking.

Furthermore, stakeholders such as healthcare professionals, therapists, and family members/partners who may work with individuals using these tools for understanding their menstrual health and wellness also experience the positive outcome of this tool. Understanding menstrual health and wellness can significantly improve one's quality of life in terms of emotional regulation, anxiety around bodily changes, or empowering people to advocate for their needs [3]. The prototyping effort involved iterative testing of the journal with potential users, which saw the meticulous and thoughtful development of layout, content, and form. The final design aims to be low-cost, accessible, and easily adaptable to different lifestyles and needs.

The iterative feedback process leveraged user interviews for primary research and secondary research through social media such as Reddit and Instagram. Menstrual tracking apps such as Flo and Clue are primarily used with a focus on data collection and cycle prediction. Wellness journals often center around mental health and productivity, but require customization to focus on menstrual health (i.e. period tracking via bullet journal pages). Some users adapt bullet journals, track data across multiple platforms, or avoid tracking altogether.

Thus, from the user-centered research, it was ascertained that people who menstruate lack a consolidated reflective, empowering, physical tool to track and understand the physical and emotional needs of their cycle. The solution, built using user-centered design methods outlined in Section 2.2, effectively addresses this gap.

2.2. User-Centered Design Methods

User-centered design (UCD) is an iterative process that places the needs of end users at the forefront of the product development process. Rather than solely relying on technology or business goals, user-centered design methods ensure that the final product is tailored to how real users interact with it. This is

accomplished through a variety of research and design techniques aimed at understanding users and involving them throughout the design lifecycle. These methods include user interviews, observations, surveys, persona creation, usability testing, prototyping, and iterative feedback lifecycles. By continuously engaging with users, designers can identify real problems, validate design decisions, and adapt quickly to solutions that users need. This focus on the user helps create a more intuitive and effective experience with the products that are developed using these methods.

User-centered design is a practice that is highly useful for a number of reasons. Firstly, these techniques enhance usability by grounding design decisions in real user behavior and feedback. UCD reduces confusion and frustration making products easier to use. Secondly, risk and cost are reduced by enabling frequent user testing to catch issues early in development. This saves a considerable amount of time and resources later in the project. Finally, user-centered design methods promote innovation and practicality. By designing to meet user challenges, grounding innovation in real-world context, unmet needs can inspire creative solutions that might otherwise have been overlooked or missed.

Several design methods were utilised to collect user feedback that informed this design process. While the bulk of user feedback was collected in the initial stage of this project's inception, the second half of the project incorporated an emphasis on user interviews and focus groups. The following outlines a brief summary of the user-centered design methods that were included in previous work as the basis for this current project.

2.3. Previous Work

In a previous project conducted by the design team to engage users in participatory design requirement identification, the Four W's framework was used to conduct initial interviews around menstrual cycle experiences. This research uncovered the following high-level themes that guided subsequent design decisions:

- Menstrual Health Tracking Tools
- Symptoms Impact Daily Life
- Gaps in [Menstrual] Information
- Inconsistency and Irregularity

In the data collection phase, eight stakeholders discussed their menstrual experiences in semi-structured interviews, which included information on coping strategies, medical interventions, tracking/monitoring, and health markers that were of interest to users. Thematic analysis and work models created from user data informed the six high-level themes and corresponding design requirements presented in Table 1.

TABLE 1: DESIGN REQUIREMENTS [1]

Themes	Design Requirements (the design should ...)
1. Own health is own responsibility	Allow the user to independently monitor and understand their health
2. Physical preparedness required	Promote physical preparedness related to one's menstrual cycle and support in planning for upcoming physical changes
3. Symptoms are normalised and dismissed	Accept and understand one's symptoms instead of normalizing and dismissing
4. Menstrual care is misunderstood and mismanaged	Provide user education on menstrual health to help users understand and manage changes
5. Menstrual cycles are unique and regularity is subjective	Embrace uniqueness by accommodating individual variability in menstrual cycle patterns
6. Convenience and unbroken routines are favoured	Seamlessly integrate into users' existing routines without disruption

The design requirement development process from past work concluded in the proposal of some user-generated ideas, including a delivery box subscription package for period resources, a visual and tangible symptom jar tracking tool, and a wall-mounted digital dashboard tracker. These concepts form the foundation for the present prototype development project.

2.4. Market Review

A competitive market analysis was performed to assess existing journals centered around menstrual health. This analysis focused on existing products, their content, and their stylistic choices, and aimed at assessing

whether a niche in the market existed for the proposed journal. This was done by querying search results on the internet and major online shopping distributors who sold journals (Amazon, Indigo, etc.).

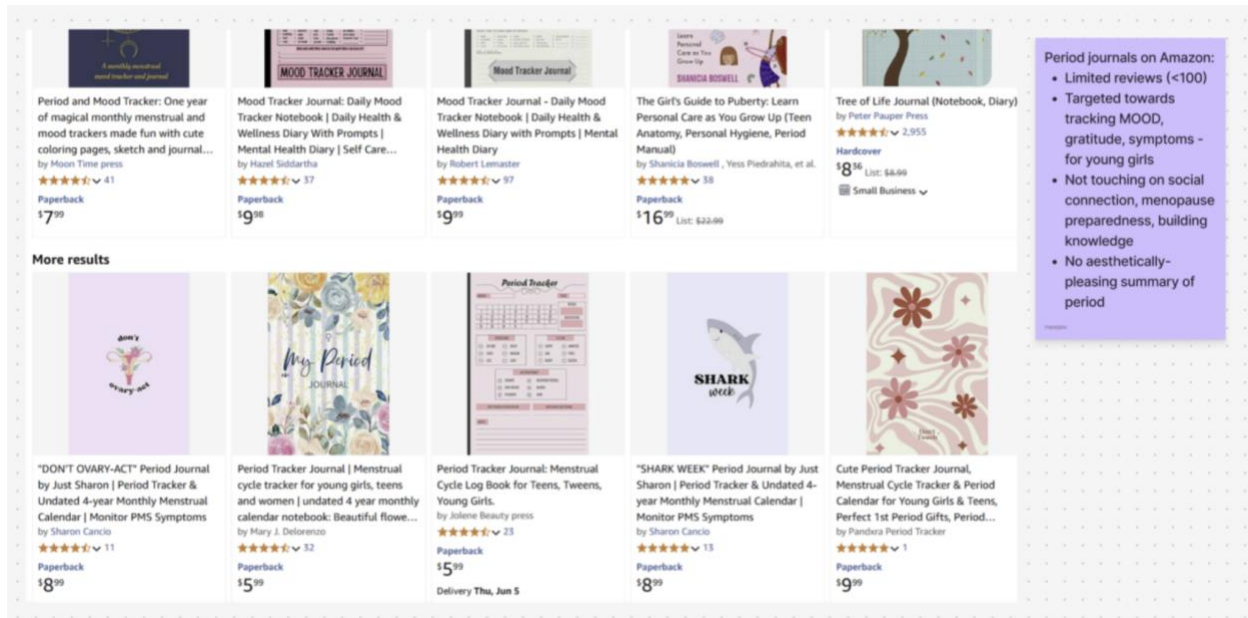


Figure 2. Competitive Market Analysis

The results of this analysis boasted a wide variety of existing products on the market (fig.2). Many journal designs featured stylistic cover art; however, menstrual health journals often displayed artwork stereotypically aimed at a female audience or aimed at younger audiences. Generic journals tended to have low user reviewer rates (despite relatively high user ratings) suggesting that few users actually engaged with the more accessible journals marketed by the seller site. Since journaling is a popular tool for reflection, it can be inferred that more aesthetically pleasing and higher-quality journals are less common or less accessible, possibly due to higher costs at niche bookstores. Additionally, their generic design meant menstrual health journals competed with general-purpose journals on mass sale platforms like Amazon, which likely contributed to their low user engagement, as they represent a niche within a niche.

Furthermore, existing journals fail to recognize that they cannot match the depth of insights provided by digital tools; instead, they should shift their focus toward offering unique, tangible experiences that complement digital tracking. Because existing journals appear not to consolidate educational content with reflection and tracking, the proposed design aims to fill this gap in the market, especially for the underserved age group of 18 to 30 year olds.

Insights from users who designed their own trackers reveal key barriers to adoption, including stylistic preferences, challenges in summarizing results, and a lack of personal connection. Additionally, traditional menstrual health journals often overlook the depth of data and interactivity offered by digital tools. To remain relevant, these journals should pivot away from trying to replicate digital functionalities and instead focus on delivering distinctive, tangible experiences that complement digital tracking rather than compete with it. The following prototype aims to address this gap in the market and improve upon existing tools to better address user needs.

3. Design Process

3.1. Requirements Definition

Previous explorations of user needs led to the prioritization of the following design requirements for the development of the present design [1]:

- 1) Allow the user to independently monitor and understand their health
- 2) Embrace uniqueness by accommodating individual variability in menstrual cycle patterns
- 3) Seamlessly integrate into users' existing routines without disruption
- 4) Promote physical preparedness related to one's menstrual cycle and support in planning for upcoming physical changes
- 5) Accept and understand one's symptoms instead of normalizing and dismissing
- 6) Provide user education on menstrual health to help users understand and manage changes

Participatory design sessions with stakeholders yielded a number of seeds from which the Menstrual Journal idea was derived. A symptom jar used to visually and abstractly represent a user's cycle inspired the physical and bigger-picture tracking elements of the Journal. Calendars and journals were proposed as mediums to promote personalization and customizability, while maintaining portability and uniqueness. Furthermore, the proposal of card-based activities and a period care basket inspired the connection and interactive elements of the journal [1]. A physical solution was favoured over a digital one to address the design requirements while proving a novel and timeless solution to the deeply personal and unique struggles around menstruation.

The Menstrual Journal aims to address an urgent, unmet need revealed by the above set of themes. There is a pressing need to foster both individual agency in understanding one's own menstrual cycle and the outspoken sharing of menstrual experiences, especially as symptoms are too often normalized and dismissed within broader cultural and medical contexts.

The Menstrual Journal is an activities diary that aims both to address cycle complexity and uniqueness while promoting meaningful engagement with community and medical professionals. The journal will include an abstract and flexible symptom tracker, a set of prompts and activities to foster deep communication, a notepad for tracking key questions or concerns, educational material on expectations throughout one's cycle, and additional features developed through iterative participatory design sessions. The initial design I0 served as a foundation for further development, guided by user engagement (Appendix, Section 6-7). The iterative prototyping process in Figma produced a high-fidelity digital prototype, which underwent user validation and culminated in the development of a pre-production physical prototype ready for pilot manufacturing.

3.2. Stakeholder Input

Stakeholder engagement was central to the *Menstrual Health & Wellness* journal's design process. In previous works, thematic analysis was conducted on multiple rounds of semi-structured interviews with stakeholders to produce the key design requirements. In the prototyping process of the journal, stakeholders engagement led to major design developments in three major phases.

Firstly, small informal interviews were conducted with a total of five people to create a harmonized prototype, I1, from some initial designs inspired by the design requirements. Generally, participants were skeptical about the value of a physical tracker compared to an app that offers analytics and convenience. However, they expressed interest in a wellness journal that could provide tangible actions, opportunities for artistic expression, and appeal to those who find comfort in journaling, prefer to disconnect from their phones, value privacy in tracking, and seek space for personal reflection. These sentiments, combined with market research, informed the development of a medium-fidelity home-printed prototype, I2

Secondly, prototype I2 was reviewed with users in semi-structured interviews to refine content and form. Users were encouraged to interact with the journal and provided in-depth feedback. The main feedback synthesized and incorporated into medium-fidelity prototype I3 can be summarized into the following points:

- Tracking sheets: Optimal number of metrics is three; Narrow wheel design switched to a bullet journal design for more flexibility in marking methods; Instructions expanded to include more example metrics; Instructions on selecting metrics, categories, and filling made more detailed.
- Reflection sheet: Optimal reflection frequency is monthly; Prompts made specific to summarizing past month and setting intentions for next one.
- Goals sheet: Goals transformed into small and engaging prompts.
- Education sheet: Technical language reduced and content shifted to body changes per phase; Definition page added to explain biological terms; Lines added on spinning wheel for readability.
- Nutrition sheets: Art and food pyramid added for visual interest; Prompt made less vague.
- Exercise sheets: Larger space created for filling plans; Prompt made less vague.
- Post card sheets: Detailed instructions page created to explain filling and mailing post cards; Option to keep post card private emphasized; Original drawings created to add more uniqueness to post cards.
- Connection sheets: Sort the prompts into categories and consider rewording some to appear more fun and inviting.
- Questions page: Make clear in the tagline that this page is for jotting down questions to ask personal support systems, and include a “purpose” column to categorize questions.
- Aesthetic: A clean look is preferred, inspired by “5 Minute Journal” with some artistic elements and a compact size.
- Arrangement: Group the learning and tracking pages first, then connection and reflection at the end. Include references to reputable sources. Remove lines from fields for user entry to allow flexibility in writing, brainstorming, or drawing.
- Other: Narrow down the audience to give the design direction.

Finally, the production-ready prototype I3 was presented to users in a focus group for final changes. They reviewed the prototype, completed a survey aimed at sparking discussion, and shared ideas together in a group in order to facilitate creativity. The focus group consisted of two individuals who menstruate, both within the target age group of 18 to 30 years. One participant was a practicing healthcare professional who was more skeptical towards the value of reflection, indicating an optimal 10% of reflection activities compared to 70% tracking and education. The other participant was a student who showed more interest in

journalling, as her desired percentage of reflection-focused activities was a higher 30%. The key feedback incorporated in the final functional prototype is summarized below:

- Tracking Sheets: The bullet tracker should be translated to landscape to make the numbers upright, as current layout necessitates rotating the page. The instructions on filling the tracker should be much more detailed: including options for how to mark the squares (e.g. stickers, filling, crosses, checkmarks), a step-by-step breakdown on choosing a metric and creating a scale, and more examples filled on the tutorial page. The metric suggestions should be more prescriptive, using the top 4 popular metrics in the example, then indicating others as additional options. Finally, duration of period should be added as a summary statistic at the bottom of every month page to allow passive tracking for reference.
- Educational Sheets: Users would benefit from the addition of a line graph showing hormone levels throughout the 4 phases, specifically the ones defined. Also add a disclaimer that cycles are variable in length on the wheel page, or at the beginning of the journal along with a mission statement.
- Goals sheet: Remove the “i want to” prompt and replace it with more prescriptive prompts relating to each topic.
- Connection sheets: Change prompt #8 to “Speculate with a friend about how public spaces can be made more period-friendly”.
- Size: The 8.5 by 6.5 inch prototype size is sufficient.
- Aesthetics: The drawings and layout are visually pleasing and the muted colour scheme is appreciated, however the introduction of more colours behind drawings will allow them to stand out more. A muted pastel watercolour theme would provide this slight increase in vibrancy. The text colour can also be changed to a dark colour that is not black.
- No changes recommended on remaining pages.

3.3. Iterative Improvements

Incorporating user feedback at each stage as mentioned above, the journal prototype went through three distinct iterations. The first proposed prototype, Iteration 0, was a collection of ideas sketched from the researcher’s past participatory design sessions and explorations of the problem space, as described in Section 3.1. Iteration 0 consisted of interactive art piece ideas, calendar dashboards, and bullet journal page sketches. The design team built Iteration 1 by proposing page designs for tracking, education, and reflection activities, using default fonts, layouts, and colours in Figma. Incorporating user feedback from semi-structured interviews, Iteration 2 saw major changes in language, layout, and content, solidifying the base for aesthetic changes. Iteration 3 was a printed medium-fidelity prototype that built on I2 by refining stylistic details, including font, colour, and artwork that emanated a minimalistic aesthetic desired by users. The final functional prototype addressed feedback that aimed at improving the usability and adoptability of the journal, including instructions, aesthetics, and physical characteristics.

4. Final Design

4.1. Prototype

The attached video provides a walkthrough of the finalized prototype, while a digital version of the prototype is found in the Appendix (Section 11).

4.2. Discussion

The final menstrual health and wellness journal integrates features shaped by user feedback, matched to core needs uncovered during the design process. A few core design decisions are presented below:

1. Independence and customisation in health monitoring: Customizability is central to the journal design, featuring user-defined tracking metrics, fillable goals, reflections tailored to individual experiences, space for personalized exercise and diet preferences, and customization options wherever possible.
2. Embracing unique cycle patterns: The interactive wheel on the Learn page and the tracking grid accommodate a range of cycle lengths. All language and guidance emphasize the natural variability of cycle symptoms and menstrual health experiences, focusing on fostering independence and understanding rather than diagnosing “irregularities.”
3. Physical preparedness and planning: Phase-specific nutrition and exercise guidance, potential symptom information, educational resources, tracking pages, and question sheets collectively encourage users to be physically prepared, plan ahead, and manage their expectations.
4. Emotional processing: Postcards and reflection prompts encourage slower, intentional writing, allowing users to gain clarity and emotional control over time.

The material and aesthetic choices also lend to achieving the core user-centered design requirements, while addressing user preferences uncovered through the iterative prototype development process:

1. Colour and illustration style: Black and white for core content with pastel watercolours and original minimal line art for illustrations allowed users to focus on relevant content while adding visual interest. This minimalistic design aligned with the style of existing general-purpose and wellness journals that users specifically mentioned liking.
2. Interactive elements: The tactile wheel, postcards, stickers, and detachable food pyramid encourage active engagement, which can result in increased memorability and a personal connection to the material.
3. Compact format of 8.5x.5.5 inches: The compact format allows users to seamlessly integrate the journal into their routines for everyday carry. The page size is still sufficiently large to allow for easy reading and writing.

Together, these design and material choices grounded in user feedback reflect the journal’s core purpose: providing a reflective, empowering, and accessible space for menstrual health and wellness.

4.3. Limitations

The *Menstrual Health & Wellness* journal design process was user-centered and meticulously documented to support the validity of its qualitative methods. However, limitations remain in terms of its universality, level of sophistication, and potential for long-term success.

Firstly, due to constraints in time, access, and scope, the diversity of the group of stakeholders involved in shaping the design was not considered. Though sweeping representation of diversity is not a requirement in qualitative research paradigms, diversity is essential to capturing user needs expertly within the design. Because personal preferences, journaling habits, menstrual health experiences, and other factors influencing user needs can vary significantly, the resulting design is limited in its universal applicability. While achieving true universality may be unrealistic for this product, future iterations should strive for greater inclusivity through more diverse participatory design, ensuring the solution continues to meet the most relevant user requirements.

Another limitation was in the sophistication of the final product, which affects both its durability and aesthetic appeal. This pilot project saw the print of a couple journal prototypes, however future iterations, following product validation, could be developed with higher fidelity depending on marketing and distribution plans. While distribution planning was beyond the scope of this project, the printable format enhances accessibility. Offering users a free, downloadable template they can print and assemble themselves promotes equitable access and allows for more frequent improved iterations at minimal cost. Future user research should explore how users weigh the benefits of a printable, accessible design against those of a more durable, professionally produced version.

The final limitation relates to the project timeline, which was confined to three months. Although user feedback was gathered through user interviews and focus groups, these evaluations were relatively short in time, giving users only an hour to form opinions on the usability of the product. Given that the journal is intended to support users over the course of a year, long-term validation is necessary to confirm its effectiveness in practice. Future research could begin with diary studies spanning one month, three months, and twelve months. These would provide deeper insights into user engagement with tracking features, the impact of reflective prompts, and the usefulness of the educational content.

Overall, this design project and resulting prototype serve as a pilot effort that requires further user-centered research before being widely deployed.

5. Future Directions

The *Menstrual Health & Wellness* journal aimed to provide users a pathway to understanding their own menstrual health, to advocate for their health, and to form connections with their community to reduce longstanding stigma around menstruation. This design endeavor culminated in a high fidelity working prototype that underwent multiple stages of user input, however it has ways to go for being distributed for widespread use. Next steps for this project are to conduct a diary study to collect situational and contextual observations, especially observing whether the journal is adopted into daily lives and whether the features

make the desired impact. The journal could potentially be distributed digitally for home printing so it may undergo tracked changes and be continuously improved with user feedback.

6. References

[1] Mouradyan, A., Pappu, N., Naufal, S., & Thompson, S. (2024, December). *From cycles to systems: Designing for menstrual health complexity* [Unpublished design paper]. University of Waterloo.

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[3] Brookssadmin, “Women’s Health Month: Understanding unique risks and prevention strategies - brooks-TLC hospital system, Inc.,” Brooks, [https://brookshospital.org/womens-health-month-understanding-unique-risks-and-prevention-strategies/#:~:text=Women%20should%20aim%20for%20at,liver%20function%20and%20mental%20health.\(accessed%20Aug.%203,%202025\).](https://brookshospital.org/womens-health-month-understanding-unique-risks-and-prevention-strategies/#:~:text=Women%20should%20aim%20for%20at,liver%20function%20and%20mental%20health.(accessed%20Aug.%203,%202025).)

7. Appendix

The appendix is attached as a separate file to this report. It contains the following information:

- Stakeholder Key
- Initial Design Requirements
- Initial Ideation
- Competitive Analysis
- Work Breakdown Structure
- Prototype 0
- Interviews 1
- Prototype 1
- Interviews 2
- Prototype 2
- Interviews 3
- Final Design